

Chapter 11

Jin Yuelin



Jin Yuelin 金岳霖 (1895–1984) was the best known analytical philosopher in China in the first half of the twentieth century. He constructed his own metaphysics and epistemology mainly by the method of logical analysis. He was among very few Chinese philosophers who were directly engaged in the discussion of some major issues in Western philosophical traditions. His philosophy doesn't belong to any Chinese philosophical school although it does adopt some Chinese philosophical terms such as “Dao” and embraces part of its traditional meaning. He took philosophical investigation as the pursuit of universal truth which is neither merely for Westerners nor the Chinese but for all humankind. Overall, he was a philosopher in China but not a philosopher within Chinese philosophical traditions. From methodology to substance, Jin's philosophy is more like those in the modern West than those in China. The main significance of his philosophy lies in bringing modern Western philosophical discourses and debates into China and participating in philosophical conversations in major Western traditions.

1 Life and Works

Jin was born in a bureaucratic family in Changsha 長沙, Hunan Province. He had six brothers and two sisters. His father was a participant of the Western Affairs Movement (*yangwu yundong* 洋務運動).¹ This might explain why Jin studied in a middle school and a high school run by Americans after receiving traditional Chinese education in elementary school. In 1911 when he was 16, he was admitted to Qinghua University, called “Qinghua School” at that time. In 1914, he went to study in the United States where he entered the University of Pennsylvania and

¹ See footnote 2 in Chap. 1.

studied business first and then political science. After receiving a bachelor's degree, he started graduate study in political science at Columbia University. He wrote a dissertation entitled "The Political Theory of Thomas Hill Green," and earned a Ph.D. in 1920. Afterward, he taught Chinese in Georgetown University for an academic year (1920–1921). He returned to China in June 1921 when his mother passed away. He went to London in December 1921, studying Philosophy for eight months, first at the London School of Economics and Political Science, and then at the University of Cambridge. He left Britain for the European Continent in September 1922 and spent the next three years in Germany, France, and Italy. He returned to China in December 1925. In fall 1926, he started teaching at Qinghua University and founded the Department of philosophy there. The first course that Jin taught there was Logic, although he himself had not studied logic systematically at that time. According to his recollection, he eventually studied logic systematically in 1931 when he spent a year at Harvard University. He taught at Qinghua University until 1952 and then at Peking University until 1955. In 1955 he was appointed as Associate Director of the Institute of Philosophy, Chinese Academy of Social Sciences and worked there until 1984 when he passed away.²

Jin's interest in philosophy was kindled in 1919 when he studied Green's thought (Jin 2010a: iv).³ However, to his own philosophical thinking, his study in Britain was most significant. By his own account, during that period Hume's *A Treatise of Human Nature* and Russell's *Principles of Mathematics* greatly influenced him. Speaking of the impact Russell's thought had on him, Jin said that Russell made him understand that precise and profound analysis may be made on those concepts often used in daily language and such analysis is philosophy. He came to emphasize analysis after reading Russell's work. As far as Hume's impact was concerned, he said that Hume was such a genius that he had to admire so much, although Hume's discussion of causality bothered him deeply (Jin 2010a: iv). Hume's problem of induction remained a central concern throughout Jin's life (Hu 2002a: 103). Jin's major works are *Logic* 邏輯 (1935), *On Dao* 論道 (1940), and *Epistemology* 知識論 (1983). *Logic* is a textbook that introduces both traditional and modern logic. It was regarded as the first book on pure logic produced in China (Liu 2004a: 349). However, Jin considered this book the worst among those three, since it contains a lot of errors. *On Dao* is the one with which he was most satisfied. It is the book in which he constructed his metaphysics. He spent the longest time on *Epistemology* (Jin 2022: 67, 69). The first draft of the book was almost finished around 1943 (Liu 2004a: 350), but he lost it because he forgot to take it with him from his hiding place after a Japanese aerial bombing was over. He finished rewriting it at the end of 1948 and sent to a publisher in Beijing. Since it was close to the time the new government took over Beijing, the publisher was no longer working as usual, and Jin's book was

²This biography is based on data from Jin (2022: 3–50), Hu (2002a: 102–103), and Liu (2004a: 343–354, 2004b: 399–432).

³In Jin's *On Dao* (2010 edition), pages numbers in both Chapter "introduction" and Chap. 1 start with page number 1. To distinguish pages numbers in its introductory chapter and in the rest of the book, I use Roman numerals for page numbers in its introductory chapter.

not published (Jin 2022: 67, 69). In 1983 when China was in the peak of thinking liberation brought by economic reform, Jin's *Epistemology* was finally and officially published 35 years after its completion.⁴ Jin's *Epistemology* was the most systematic discussion of theory of knowledge ever in China. It presents Jin's own views of knowledge and was an attempt to continue conversations on knowledge in Western philosophy. Regardless of how much its critique of Western epistemology and its effort to establish a new theory of knowledge succeeded, it was a significant work for modernization of philosophy in China.

2 Philosophy, Logical Analysis, and Metaphysics of *Dao*

To understand Jin's philosophy, we may need to first know his view of the nature of philosophy. According to him, philosophy is a prejudice that gives a certain reason (說出一個道理的成見). The reason for him to say so is that the most fundamental part or starting point of a philosophy is always out of an assumption or faith rather than from reason. Such a starting point of a philosophy is either temporary or permanently unprovable or unfalsified. According to him, if one wants only to use a starting point of their philosophical system after first proving it, they would never be able to establish their philosophy. Philosophers always construct their philosophies from presupposed assumptions or fundamental biases. What assumptions they have chosen depended on their own personality and their times and environment (Jin 2009: 459; Wang 1993: 74). Given such an understanding of philosophy, it seemed to Jin that inference is the key to constructing and expressing philosophy. Greatly influenced by logical positivism, Jin believed that logical analysis is crucial for philosophy.

If philosophy is mainly about argumentation, then logic is the essence of philosophy. ... Very often philosophers are criticized not because of their thought but because of their method of developing their thought, and many philosophical systems are destroyed by hitting the reef of logic. (quoted by Wang 1993: 76)⁵

Jin's philosophy may mainly be divided into metaphysics and epistemology, while the former is more fundamental than the latter (Hu 2002a: 103). Although Jin embraced logical analysis as the basic method for philosophy, he insisted that metaphysics, whose fundamental concepts go beyond logic, is ultimately important for philosophy. Although the fundamental assumptions in metaphysics are not logically provable, intellectually and emotionally we need to discuss them.

We must remember that "metaphysics" is a good term that means what is higher than or transcends physical things or natural things. ... Here this term refers to a branch of

⁴After 1949 when Marxist philosophy became the state ideology in China, Jin's philosophy was not officially acceptable. His *Epistemology* was unofficially published in 1958 as a reference for the purpose of criticism (Liu 2004a: 351).

⁵In this chapter, all translations are mine unless indicated otherwise.

philosophy. It discusses those fundamental ideas and concepts that can't be proved or refuted. It is such an area in which our test and analysis of assumptions, postulates, hypotheses, and basic premises or those that we randomly name offer us some options, which can serve as the starting point for any philosophical discussion. (quoted by Wang 1993: 79–80)

Some basic premises and fundamental conclusions in metaphysics are beyond rational justification, but we as human beings desire them. Speaking of his own motivation for the study of metaphysics, Jin said: "... I pursued not only intellectual understanding of studied objects, but also emotional satisfaction from the outcome of studies" (Jin 2010a: xviii; Wang 1993: 80). For Jin, although logical analysis alone can't build a metaphysics, it can significantly improve traditional metaphysics. He acknowledged that logic and epistemology are weak in traditional Chinese philosophy, which needs to be greatly developed (Jin 1998a: 274–275). However, as a Chinese, his attachment to Chinese philosophy made him do as much as he could to give a Chinese flavor to his philosophy that is fundamentally Western as far as its issues and approach are concerned. To a great degree, his metaphysics is an attempt to develop some existing metaphysical theories in Western philosophy with redefined terminologies in traditional Chinese philosophy. Also, it is worth noting that Jin was greatly motivated to create a new metaphysics to help solve Hume's problem of induction, since he believed that Hume's challenges to the law of causality and induction were caused by flaws in his entire philosophical system, while a new metaphysics may fundamentally demonstrate the law of causality as a necessary connection between things and the universal effectiveness of induction (Jin 2010a: iv–v).

For his own metaphysics, among others, Jin first adopted the term "Dao 道" (Way) from traditional Chinese philosophy and made it the highest concept of his metaphysics. The reason for his doing so is to connect his philosophy with traditional Chinese philosophy, showing his attachment to Chinese philosophy. As he put it, he used "Dao" because "Dao has Chinese flavor." "Dao" has been the highest concept for many Chinese philosophers in history, one of the most representative terms in Chinese philosophy. To bring "Dao" into his metaphysics, Jin certainly gave his system of philosophy some Chinese characteristics. Nevertheless, putting "Dao" in his metaphysics is not required by his reasoning but by his emotional need. As Jin admitted when he talked about his *On Dao*: "Intellectually, I may use another term instead of so-called 'Dao,' and doing so doesn't affect the thought of this book; but emotionally, a different term other than 'Dao' probably will not touch my heart, please my feeling, and cultivate my nature" (Jin 2010a: xviii). Jin's use of "Dao" in his metaphysics is like "putting new wine into an old bottle," but not completely so. After all, the meaning he gave to "Dao" overlaps its meaning in traditional Chinese philosophy to a certain degree. In different Chinese philosophical schools, the meaning of "Dao" is different. For example, for Daoism, "Dao" means the natural order and the ultimate sources of all, while in Confucianism "Dao" refers more to the right way or the highest truth of the universe and human life. In Jin's metaphysics, "Dao" is similar to some traditional Chinese understandings in the following sense: (1) the *Dao* is the ultimate substance or the original reality (*shiti* 實體 or

benti 本體), (2) the *Dao* is the entire changing process of the world, and (3) the *Dao* is the general law of the world (Wang 1993: 89–91). However, unlike any concept of the *Dao* in the Chinese philosophical tradition, “*Dao*” in Jin’s metaphysics could be explained and analyzed into “energy” (*neng* 能) and “form” (*shi* 式) and defined as “*shi-neng*” or “the combination of *shi* and *neng*” (Jin 2010a: 22). It is such a unity as the original reality that is the source of all things and ultimately real. Logically, the *Dao* as a transcendental world of *shi* and *neng* which is not in space and time produces the natural or actual world by its activity. Everything in our perceivable world is merely the expression and individualization of the world of *shi-neng* (Hu 2004: 91). To distinguish better what is in space and time and what is not, Jin reserved “existence” (*cunzai* 存在) only for that which occupies space and is in time. For that which is in the world of *shi-neng*, its presence can only be called “realness” (*shizai* 實在).

Neng (能 *Energy*) and *Shi* (式 *Form*)

To understand Jin’s metaphysics, we obviously need to know his notions of *neng* and *shi* more. According to him, “*neng*” and “*shi*” are similar to Aristotle’s “matter” and “form” on the one hand, and similar to Zhu Xi’s “material force” or “vital energy” (*qi* 氣) and “principle” (*li* 理) on the other (Jin 2010a: xiv). For Aristotle, everything is the unity of matter and form, while for Zhu Xi 朱熹 everything is the combination of *qi* and *li* where *qi* is *stuff* and *li* is pattern. Similarly, for Jin, everything is made of *neng* and *shi*. However, differences between Jin’s notions of *neng* and *shi* and Aristotle’s notions of matter and form, as well as between Jin’s notions of *neng* and *shi* and Zhu Xi’s notions of *qi* and *li* are significant.

For Jin, *neng* doesn’t have any particular form or property. “*Neng*” is a proper name like a person’s name in the sense that it doesn’t have any predicate (Jin 2010a: 2). Unlike a class name such as “water” or “electricity,” “*neng*” doesn’t refer to a type of thing. To illustrate what *neng* is, he used this example: when a cigarette was finished, it became ash, then ash becomes a small unit of stuff. After continuous dissolution of ash, at last there is still something left as the most basic matter that has no describable attribute, that which serves as the common base for all things. That is what Jin called “*neng*.” In his words, when a cigarette is gone and becomes ash, “*neng*” exits from cigarette and enters ash; when ash merges into soil and no longer exists as ash, “*neng*” exits from “ash” and enters “soil.” *Neng* doesn’t cease to be but merely enters different forms of things (Jin 2010a: 2). For Jin, *neng* is pure matter but transcends space and time. The word “*neng*” is used for the sake of convenience in expression but doesn’t imply any attribute. Because of its lack of any describable attribute, *neng* is inexpressible in language and can only be known by intuition or experience defined in a broader sense (i.e., the experience that includes thinking procedure) (Jin 2010a: 2–3). It seems that Jin asserted *neng* by inference. At this point, we may see that *neng* is not the same as matter in Aristotle since Aristotle’s matter in general can have properties. For example, brick as matter of

house certainly has certain properties. Furthermore, for Aristotle, the difference between matter and form is relative. Matter for a being at a higher level could be form for a being at a lower level.⁶ In Aristotle's metaphysics, only pure matter at the lowest end of the hierarchy of matter has no describable attribute.⁷ But pure matter is not the one that combines with form. Regardless of *neng*'s being indescribable, Jin did say something to explain what *neng* is. According to him, unlike Aristotle's matter, *neng* is active and dynamic but not passive and static (Jin 2010: 3). In Aristotle's metaphysics, form is active and matter is passive since form functions as causes of motivation and purpose for matter. In Jin's system, *neng* is self-motivated and self-caused. In this aspect, *neng* is more similar to *qi*, which is dynamic in Zhu Xi's metaphysics, but *qi* in Zhu Xi is not pure matter that has no property. Furthermore, for Zhu Xi, logically, *li* is prior to *qi*. Although everything is a unity of *qi* and *li*, it is *li* that determines what a thing is. In Chinese philosophical traditions, there was a concept of "pure energy" or "pure material force" (*zhengqi* 真氣) that is beyond description. Feng Youlan developed the concept and claimed that pure vital energy is merely a logical concept that doesn't exist in the actual and perceivable world.⁸ It seems that Jin's concept of *neng* overlaps with that of *zhengqi* which transcends time and space but Jin's *neng* does have the ability of movement in different parts of *shi* which is also not in time and space. *Neng*'s characteristics will be explained better once we know more about what Jin meant by *shi*.

According to Jin's definition, *shi* is the possibility of the infinite disjunction of the total of all possibilities [i.e., the possibility of possibility (1), or (2), or (3), or (4), or (5) ... or (∞)] (Jin 2010a: 5). What is logical is what is possible. Each possibility is a mold or frame which may actually have contained or not contained *neng* but could contain *neng*. Then, *shi* is the mold or frame for the infinite disjunction of molds or frames. In this sense, we may regard *shi* actually as a total of all possibilities, and any possibility is a part of *shi*. On the one hand, *shi* is similar to Plato's world of forms or ideas which is not in the perceivable world but provides all possible molds for things to be realized in the actual world. On the other hand, each possibility in *shi* is similar to an Aristotelean form in the sense that it defines the essential nature of each actual thing. But how can *neng* and possibility get together to bring actuality? According to Jin, *neng* may freely enter or exit from one or multiple possibilities. Those possibilities that have not had *neng* inside are merely empty concepts or empty molds, while those possibilities that have had *neng* inside are actualized or become "realized possibilities" (*xianshi de keneng* 現實的可能). They are what Jin defined as universals (*gongxiang* 共相). According to Jin, the universal is the realized possibility, "the frame or mold that already has had *neng* inside" (Jin 2010a: 3–4). Universals are realities, not concepts in people's minds. A universal is also the possibility of individualization (*geti hua de keneng xing* 個體

⁶For a brief discussion of this difference between Jin's *neng* and *shi* and Aristotle's matter and form, see Wang (1993: 93–94).

⁷Whether Aristotle indeed believed in such pure matter has been questioned. See Hugh R. King (1956).

⁸See Chap. 10.

化的可能性) or the possibility to be expressed or shared by many individuals. In other words, it is a realized possibility of individualization. It means that a universal must be shared and expressed by many individuals in space and time in the actual world (Jin 2010a: 3–4). Those empty concepts such as “superman” or “dragon” as possibilities have not become real universals since *neng* has not entered them and there are not particular things that express them, but logically they could become universals. Therefore, they are possibilities that may have *neng* but don’t actually have *neng* (Jin 2010: 3–4). In short, the world of *shi-neng* is logically prior to all individuals in the actual world. Here is how it is related to the actual world in a logical sense: when *neng* enters certain possibilities in *shi*, some universals become actualized, which bring about various kinds of individuals in space and time and then the world in which we live occurs, which is the natural world or what Jin called “the world as it is” (*benran shijie* 本然世界) (Guo 2006: 441).

According to Jin, *neng* and *shi* are inseparable. *Neng* can’t escape from *shi* or be outside *shi*. In other words, *neng* must be in some possibilities. Therefore, there is no *neng* without *shi*. Similarly, there is no *shi* without *neng* (Jin 2010a: 7–8). Although there are some possibilities as empty concepts without *neng*, *shi* as the possibility or mold of the disjunction of all possibilities must have *neng* in some of possibilities listed in the disjunction. Therefore, although there are possibilities or molds that don’t actually have *neng*, overall there is no *shi* without *neng*. Given his definition of *shi*, all possibilities are within *shi*, “... that is to say, there is no possibility that is not listed in *shi*.” Then, we can easily know that “outside *shi* there is no mold or frame for *neng*” (Jin 2010a: 7). Then, *neng* can only be in *shi*. Since *neng* can’t be destroyed, in *shi* there always is *neng*. This certainly means that *shi* can’t be without *neng* and *neng* can’t be without *shi* (Jin 2010a: 7).

Clearly, to make *shi* the frame for disjunctions of all possibilities, Jin was trying to avoid Aristotle’s admission of pure matter without form or pure form without matter. However, if Jin directly had named each possibility “form,” he would have to conclude that there are some forms without *neng* or matter. Therefore, by defining “*shi*” differently from Aristotle’s “form,” Jin didn’t totally succeed in avoiding Aristotle’s problem (i.e., the separation between matter and form in the case of pure matter and pure form). Since Jin’s “possibility” is similar to Aristotle’s “form,” by asserting that some possibilities lack *neng*, at least he has to admit that there are some “forms” without matter. Of course, these pure forms are not the same as Aristotle’s in the sense that they are not at the top of the hierarchy of all forms and that they are not the only ones that have no matter. In Jin’s metaphysics, the separation between possibilities and *neng* is not exceptional but common. Those possibilities that don’t have *neng* are part of *shi* and exist independent of human minds. Furthermore, *shi* as the combination of empty concepts or molds that have no *neng* and universals that are expressed by particulars formed by *neng*’s entering certain possibilities is not in time and space but real and outside epistemic agents, which is

close to Plato's world of ideas. Therefore, Jin's emphasis on the inseparability of *neng* and *shi* doesn't prevent him from falling into metaphysical idealism.⁹

According to Jin, neither *shi* nor *neng* in themselves are in space and time and are not subject to change, but individual things as expressions of certain universals produced by the combination of *neng* and particular possibilities do have a beginning and end, or life or death, and are in constant change. When *neng* comes in or out from a particular possibility, a particular thing or particular kind of thing may come to exist or extinguish, but *neng* and *shi* in themselves are not affected. For example, when a person is born, it merely means that *neng* enters the possibility of a living particular person, while when a person dies, *neng* comes in the possibility of the person's being deceased. Similarly, when a type of thing comes to exist or dies out, that is also about *neng*'s coming in or out from certain possibilities. For example, when there were still dinosaurs, *neng* entered the possibility of dinosaurs, and when dinosaurs became extinct, *neng* exited from the possibility of dinosaurs. When new things occur, it means that *neng* enters certain possibilities that it didn't before. For instance, before airplane as a kind was invented, *neng* didn't enter the possibility of airplane, and it entered the possibility when airplanes came to exist and stays there as long as airplanes continue to exist (Jin 2010a: 24). His explanation of the birth and death of species or particular kinds of thing as such is certainly consistent with his claim that when *neng* enters a possibility a universal comes to be. But his explanation of the birth and death of a particular individual in terms of *neng*'s entrance or withdrawal from a possibility doesn't fit his view that a realized possibility is a universal but not an individual in a kind. Given that *shi* includes all possibilities and *neng* only enters or exit from possibilities, all birth and death or beginning and end happens within *shi* and is determined by *neng*'s activity. *Shi* and *neng* in themselves will never change or de cease. Their expressions via individual things are in space and time but they themselves are not. Similarly, the *Dao* as the world of *shi-neng* has no beginning and end either (Jin 2010a: 12–14). Nevertheless, Jin also claimed that *shi-neng* can't be totally separated from time and space in the sense that universals produced by the combination of *neng* and possibilities must be expressed by individuals in space and time. A universal is not dependent on any particular individual to be expressed but does need at least one individual to do so. For example, if there is not even one red thing, redness as a universal will no longer be real.

Unlike the traditional Chinese understanding of the process of change in the world, in Jin's metaphysics, the *Dao*'s movement is progressive but not regressive or circular (Wang 1993: 88). Jin named the process of the *Dao*'s movement "from wuji (無極) to taiji (太極)." "Wuji," which literally means "without the utmost point," is about infinite earlier point of the *Dao*. Since the *Dao* doesn't have a beginning, a definite starting point of the *Dao* doesn't exist, Jin referred to its infinite past as "wuji." Jin also interpreted it as "chaos" in which there is almost no possibility that has had *neng* inside. "Taiji," which literally means "the ultimate one," is the *Dao*'s infinite future. Since the *Dao* doesn't have an end and there is not an eventual ending

⁹For a similar critique of Jin's view see Xu et al. (1992: 423).

point of the *Dao*, Jin referred to the infinite future point as “taiji.” At *taiji*, *neng* entered or enters almost all possibilities (*neng* exited from those possibilities which are related to those extinguished species or things) (Jin 2010a: 195–198, 209; Hu 2004: 53). Also, at *taiji*, the highest level of truth, goodness, and beauty are achieved and form a unity (Jin 2010a: 211–212). Clearly, by talking about “from *wuji* to *taiji*,” Jin indicated that the *Dao* is in movement with a forward and progressive direction. Given that the *Dao* determines the actual world and manifests in it, the actual world must move toward the same direction as the *Dao*. It seems that this part of Jin’s metaphysics greatly reflects his emotional side without logical argumentation. Actually, it contradicts his logical reason, since progressive movement of the *Dao* in the transcendental world makes no logical sense. However, if such a movement is not in his transcendental world of the *Dao* as the unity of *shi* and *neng*, progressive movement in the actual world that he firmly believed becomes groundless.

Although Jin’s metaphysics as such is very interesting and original to some extent, it has more problems than what is mentioned above. For example, it seems illogical to assume that there are containers (possibilities) which accept *neng* as pure matter and *neng* that moves in or out from one container to another in a non-spatial-temporal world. If a possibility is a mold or frame for *neng* to enter or exit, it needs to be in space and time; *neng*’s entering or exiting from one mold to another must involve time and space. If possibilities in *shi* are purely logical, it doesn’t make sense to consider them containers that *neng* can enter or exit from. Jin seemed to realize this problem. To deal with it, Jin emphasized that all *neng*’s activities are not like entering or exiting from one place to another in an ordinary sense. Its movement is not like the movement of an ordinary moving thing (Jin 2010a: 20–21). Therefore, by *neng*’s entering or exiting from possibilities, he seemed to metaphorically talk about the combination or separation of *neng* and possibilities but didn’t literally mean the existence of containers (possibilities) and the moving agent (*neng*) in non-spatial-tempo world. But Jin didn’t explain what he exactly means by the movement of *neng*. He emphasized that “*neng* can enter and exit” is very important in his metaphysics (2010a: 20), but never explained what *neng*’s entering and existing are like if they have nothing to do with movement in space and time. In his metaphysics, *neng*’s movement serves as a key explanation of realization of universals and changes in the actual world, but it is unclear what it is like. Furthermore, unlike passive matter in Aristotle’s metaphysics, Jin’s *neng* obviously is an active agent. That isn’t consistent with his claim that *neng* has no describable attribute, as pointed out by Hu Jun (Hu 2004: 59).

Relatedness of Universals and Universal Laws of Nature

Jin firmly believed in the uniformity of nature. To defend such a belief, he needed to show that there are the universal laws that entail necessary connections between things in the actual world, which is partly his solution to Hume’s problem of induction. The way for him to justify his claims on the existence of universal laws is

closely associated with his understanding of the universals and relatedness of universals. Before examining his relevant arguments, we need to further clarify his conceptions of universal, particular, and individual. As mentioned earlier, for Jin, a universal is a realized possibility, i.e., a possibility in which *neng* enters. It is actually the realization of a possibility. For example, when *neng* enters the possibility of redness, the possibility of redness is realized and the universal “redness” comes to be real. Meanwhile, a realized possibility or universal is also “the possibility of individualization,” i.e., the possibility of having many individuals. Given that a universal is already a realized possibility, it must be a realized possibility of individualization, which means that it is shared and expressed by many individual things. For example, when the universal “redness” becomes real, many red objects come to exist. The realness of “redness” necessarily means the existence of multiple red things.

An unrealized possibility doesn't have concrete and individual expression, which is not a universal, since so-called “universal” is something shared by some individuals, while an unrealized possibility, because of its unrealization and lack of ability of concretization and individualization, in itself is not a contrast to the individual, and therefore can't be spoken of as “universal.” (Jin 2010a: 60)

On the one hand, universals are real although they are not in space and time, and on the other, universals are not disconnected to time and space and individuals, since they are necessarily manifested into individuals in space and time. These individuals express universals in their unique way. For example, redness must be manifested in different red individuals. The unique redness in a red individual is a particular of redness. It is worth noting that Jin is very clear about the distinction between particular and individual: the former is merely a unique aspect of a thing (e.g., a special redness), while the latter is a concrete thing or a complicated whole that has various properties and relationships which express multiple universals (Hu 2002a: 114–115). For instance, a particular Maying tree as an individual object expresses universals in physical, chemical, biological, color, and form aspects, etc. (Hu 2004: 100).

Universals in the actual world are connected with each other in certain ways. First, universals in a certain aspect such as a chemical or physical aspect are connected. Such connections or orders are straightforward (*zhixian de* 直綫的) and not turning back (*bu huitou de* 不回頭的). They are following order logically (Jin 2010a: 96–97; Hu 2004: 100). The relatedness of universals in the same aspect forms an internal group. Each science pursues knowledge of the connection between universals in a particular aspect. Second, universals in different aspects are also connected (Jin 2010a: 94–96). Since universals in any aspect are not independent of universals in other aspects, they can't be independent of connections of universals in other aspects. Therefore, they are also connected with universals in other aspects. Connections of universals are also at different layers. Furthermore, the relatedness of universals is never conflicting but always consistent, since the world of universals follows logic (Hu 2004: 106). Given connections among universals as such, all universals are connected together and form a whole coherent picture (*tuan* 圖案) which is the *Dao* as a total. Meanwhile, each universal is a *dao* as well (Hu 2004: 101). Importantly, for Jin, connections between universals are certain (*yiding de* 一定的) which has no exception. He explained as follows:

Here so-called “certain” means no exception. Please note that the relatedness of universals of course has no exception, and if there is exception it will not be a connection between universals. The reason why it is called “certain” but not “necessity” here is a matter of habit of wording. I personally like to restrict the use “necessity” to necessity of pure principle (*chunli* 纯理).¹⁰ (Jin 2010a: 145–146)

Obviously, for Jin, relatedness of universals is what is commonly called “necessary connection.” Connections as such are objective and unchangeable orders or laws in the actual world (Hu 2004: 101). Therefore, if relatedness of universals indeed is necessary connections and universal laws that allow no exception to exist, then the uniformity of nature must be real. Although this is an extremely important point in Jin’s metaphysics, Jin didn’t make explicit argumentation (if any) to show that relatedness of universals is necessary or certain. It seems that Jin assumed it with his belief in the order of the world without proof.

To explain changes in the realized world or actual world in which we live, Jin attempted to form a theory of the relation between necessity and contingency in terms of relatedness of universals and changes in individuals. To illustrate this, Jin brought *li* (理 principle) and *shi* (勢 tendency) as a pair from traditional Chinese philosophy into his metaphysics and gave them a new interpretation. In the context of his system, “li” no longer means “principle” but “relatedness of universals,” while “shi” no longer means “tendency” but “arising and ceasing of a particular individual in a particular time and space.” *Li* or connections between universals such as the one between cause and effect are certain and unexceptional. Individuals can’t violate *li* to which they are subject. However, when and in what specific way a particular individual realizes *li* or a certain connection between universals is accidental. This explains why an individual changes at a certain time and space, including its birth and death, are not pre-determined and predictable. He proposed that in individuals’ changes “*li* is intrinsic and inevitable but *shi* is not necessarily realized” (*li you gu ran, shi wu bi zhi* 理有固然, 勢無必至). How a specific causal relation affects an individual may be a clear case of this. He argued that a real causal relation always exists although it might not be realized when its necessary conditions are not satisfied. For example, suppose one took a fatal poison that would certainly cause his death without timely interference, but his doctor gave him a medicine that made him throw up the poison before it affects his body and therefore he survived. The causal relation between the poison and death of a person is still real but it is not realized due to the intervention of a medicine which makes conditions of that causality unsatisfied. Meanwhile, the causal relation between medicine and poison is realized—the medicine caused the poison to be ineffective (Jin 2010b: 500).

Nevertheless, Jin still needed to explain better what happens when a particular thing comes to exist or perish. As mentioned earlier, Jin used *neng*’s entering or exiting to explain individuals’ birth and death at a particular time and space. According to him, when a particular thing comes to be, *neng* entered the possibility of the particular thing, and when a particular thing dies *neng* exited from the

¹⁰ By “pure principle,” Jin means the proposition that only expresses the relations of ideas and whose truth is necessary.

possibility of the particular thing. To illustrate this further, he used other two terms in traditional Chinese philosophy, “ji” (几) and “shu” (數), with some degree of new interpretation. According to him, “ji” refers to the moment when *neng* is about to enter a possibility or to exit from a possibility, while “shu” refers to *neng*’s capability of entering or exiting from a possibility. Whether something could happen is determined by *shu*, while exactly when something happens is up to *ji*. For example, a person’s eventual death is *shu*, but exactly when he will die is *ji*. In a specific situation, if it is known that one will die tonight but not certain exactly when tonight he will die, his death tonight is *shu*, but the exact time of his death tonight is a matter of *ji*. In short, *ji* is unpredictable but *shu* is not unknown (Xu et al. 1992: 427–428). However, his theory as such can only explain why a particular individual has a certain essential quality in term of universals that are realized when *neng* enters the possibility of the individual but fails to show why the individual has something unique given that *neng* is the same in all possibilities and the combination of *neng* and possibilities can only produce universals.

For Jin, his metaphysics based on *neng* and *shi* has asserted universal orders and objective laws in our world. Necessary connections between universals or *li* as natural laws guarantee the uniformity of nature or the similarity of the past and the future which Hume is skeptical about and refutes Hume’s denial of causality as necessary connections. Although Jin is correct in seeing that if there are universally valid laws in nature and logical order in the actual world Hume’s challenges to causality and induction would certainly be met, empiricists such as Hume wouldn’t accept any of Jin’s assumptions about necessary connections between universals and universal laws in nature. For Jin, the reason why Hume couldn’t acknowledge universal laws and orders in nature is that he doesn’t admit the existence of abstract concepts and considers “ideas” something that are only based on impressions which are always associated with images (*xiang* 像).¹¹ From such a concept of idea, Hume couldn’t find what is universal and therefore fails to realize objective and universal orders in nature (Jin 2010a: iv–v and 2010b: 306–307; Hu 2004: 9–10).

3 Theory of Knowledge

Jin’s epistemology is an attempt to continue conversations on the major issues in the theory of knowledge in Western philosophy and has nothing to do with traditional Chinese philosophy. For him, the theory of knowledge aims to find universal principles of knowledge and therefore must be able to establish objective criteria of knowledge. Overall, Jin’s theory of knowledge may be classified under both realism and reflectionism. It presupposes the existence of external things and takes sensory experience as the starting point of epistemic process. Furthermore, it asserts that

¹¹ Jin seemed to overlook that impressions discussed by Hume are much broader than those of visual images. This point is made by Lawry Finsen in his comments on the earlier draft of this chapter.

senses can correctly reflect what is sensed when normal sense organs of an epistemic agent function appropriately. Moreover, truths of knowledge claims lie in the correspondence between propositions and facts. All major conclusions in his epistemology seem to be so ordinary and pre-modern that they all could be found in classical epistemology prior to Descartes. However, Jin didn't simply return to traditional views but provided them with a modern defense by his epistemological system.

The Starting Point of Epistemology and Critique of Subjectivism in Modern Epistemology

According to Jin, like metaphysics, epistemology must also start from something unprovable. He called the starting point of epistemology the “starting mode” (出發方式) that consists of starting materials and starting principles (Jin 1998b: 31). Clearly, not all assumptions as starting points are equally good for building a theory of knowledge. His effort to build his epistemology starts with the critique of the dominant trend of epistemology in modern Western philosophy. According to him, starting from the assertion of the existence of external things is much more plausible than to doubt, deny the existence of external things, or infer the existence of external things through the subject (the epistemic agent). Why? Because objective and universal knowledge are possible only if there are external things. Jin held that a tenable epistemology that is able to find universal truths presupposes the existence of the external world. Jin was well aware of the major debates on the knowledge of the external world in modern Western philosophy and acknowledged that starting from the subject or epistemic agent who has thought or sensory experience is the mainstream in modern epistemology. For him, such a starting mode in epistemology is actually starting from a particular sensory experience in a special time and space as its starting material and starting from the principle of indubitability as its starting principle. He named such a starting mode as a “mode of subjectivism 唯主方式” (Jin 1998b: 35). Those philosophers who hold this type of starting mode all start from sensory experiences of the epistemic agent, no matter whether they deny or doubt the existence of the external world, or indirectly admit the existence of the external world through inference. According to Jin, all those epistemologies that adopt the mode of subjectivism had failed to establish a tenable epistemology. Why? He offered two closely connected reasons. First, such epistemologies couldn't succeed in establishing the existence of the external world, even if some of them attempted to do so. Second, those epistemologies couldn't establish universal and objective truths of knowledge (Jin 1998b: 44–45).

Why can't the mode of subjectivism in epistemology establish the existence of external objects? Why is the failure in establishing the existence of external objects a problem for epistemology? According to Jin, among those epistemologists who start their theories from the sensory experience of the epistemic agent, some indeed intended to establish the existence of external objects, but they attempted to do so

through the content of sensory experience. Such an effort can't succeed since from what perceivers can perceive one can't necessarily conclude that there must be external things that cause their perception or sensation. Why is establishing the existence of external objects indispensable for a tenable epistemology? For Jin, without external things as objects of knowledge, objective and universal knowledge will be impossible. If there are no objective and external objects, there will not be any objective and universal criteria of knowing and knowledge will become a matter of subjective options, which is contradictory to what knowledge is about. For Jin, knowledge must be objective and universal, and so not dependent on individuals' subjective opinions. Therefore, failing in establishing the existence of external things in epistemology is to fail to explain what knowledge is. For him, if an epistemology has no objective foundation and fails to explain true knowledge, it will have no sense of authenticity.

The sense of authenticity of a theory of knowledge depends on the reality of objects of knowledge. The reality of objects of knowledge is guaranteed by the existence of external things. According to Jin, external things are characterized by the following: (1) their existence is independent of our sensory experience; (2) their qualities are independent of their perceivers or epistemic agents; and (3) they have continuous identities. For example, a painting is the same object before and after we bought it (Jin 2010b: 79).

As far as the starting mode of his own epistemology is concerned, Jin held that starting materials are both the existence of external things and sensory experience, and the starting principle is that of effectiveness. The principle of effectiveness is the one that can explain and accept these characteristics of external things mentioned above (Jin 2010b: 84). Jin is reluctant to label his own epistemology with any "ism" such as materialism, idealism, empiricism, or rationalism. However, he said, if he had to choose a label for his epistemology, it would be "realism" (實在主義). Jin's realism was greatly influenced by early works of Bertrand Russell and G. E. Moore, although he disagreed with Russell's later turn toward less assertion of the external world. Jin admitted that he just directly asserted the existence of the external world without proof except appealing to common sense. In that aspect, he was obviously following Moore. For Jin, the existence of the external world can't be inferred from what the subject can sense or perceive. He insisted that an epistemology that can demonstrate the possibility of objective and universal knowledge must be based on the assertion of the existence of external things, although the existence of external things is not provable. After all, we all actually believe that the external world does exist no matter whether philosophically we hold such a belief or not. For example, we know for certain that our mothers exist externally to us. We know that our friends in front of us are real. We don't infer their existence from our senses. We don't need to prove it through any sophisticated theory. Our common sense of it is very basic so that it can be called "instinctive faith" (*benneng de xinyang* 本能的信仰). Not all instinctive faiths are reliable, but those unreliable instinctive faiths can only be ultimately overthrown by other instinctive faiths (Jin 2010b: 54). A tenable epistemology should accord with our common sense that admits the existence of the external world and even takes it for granted. Therefore, he simply asserts the existence of the external world and makes it the starting point of his epistemology.

Sensory Experience, Ideas, and Knowledge

Granted that the external world is real and that we perceive it via our senses, then we need to find out whether we can know the external world in the way it really is and what process we must go through to know the external world. For Jin, we can certainly know the world in the way it really is. How can we know that? His answer again is common sense. For him, we actually believe that we can know the world in the way it really is. There are two steps for us to acquire knowledge. The first step of our epistemic process is to obtain materials of knowledge by correctly using our senses. The second step is to obtain knowledge that is universal and abstract based on materials of knowledge that are always particular and concrete.

The first step of knowing relies on our senses. When we use our senses correctly, our sensory experience will reflect the way the external world really is in the sense that it may either perceive some external objects in their entirety or partiality. When a normal person with normal senses uses her senses correctly, what results is called “correct sense perception” (*zhengjue* 正覺) (Jin 2010b: 92). *Zhengjue* is different from illusions that may appear in our dreams or abnormal mental states. What is presented in *zhengjue* is its content or “presentation” (*chengxian* 呈現). Sensory activities usually have contents. In the case of *zhengjue*, the content is called “given” (*suoyu* 所與). Here is what he said about *suoyu*:

Suoyu is either an external thing or part of an external thing. There are two aspects of *suoyu*’s position: it is content and object at the same time; as far as being content is concerned, it is presentation, while as object, it is the external thing or part of the external thing with objectivity (*dui xiang xing* 對象性). In correct sense perception, *suoyu* unites content and object as the one, although on other occasions, content and object may not be the one. For example, when I was imagining my friend in London, the content of my imagination was one thing, while the object of my imagination was another thing. *Suoyu* as content comes and goes with the activity of senses; as object it exists independently of the activity of senses. (Jin 2010b: 96)

Notably, Jin insists that *suoyu* as the content of *zhengjue* is not something caused by the external thing or something representing the external thing but the external object itself. Logically, this doesn’t make good sense. What is presented in one’s mind and what exists externally to one’s mind can’t be exactly the same thing, although it is logically possible that the former reflects exactly what the latter is. Moreover, as a critic of Jin has pointed out, *suoyu* as “presentation” or the content of *zhengjue* must have been conceptually processed by the epistemic agent, which makes it impossible to be an external object independent of the mind (Chen 2021b). It seems that by saying that *suoyu* is both the content of sense perception and the external thing Jin was trying to emphasize that in the case of *zhengjue* what is reflected in one’s sensory experience exactly matches with the external object that one has sensed. If so, it would be more accurate for Jin to say something like “*suoyu* as the content of *zhengjue* presents exactly what the external thing is.”

Jin’s belief that *zhengjue* can reflect exactly the way the external thing is is closely associated with his view that in *zhengjue* what one directly senses is the external object itself but not what the external thing causes or represents in the

mind. He argued that the relation that *zhengjue* involves is only between the individual who has sense perception and the external thing that is sensed, but not between the presentation and the external thing. In other kinds of sensations, the external thing is regarded as the cause of presentation and the presentation is the effect of the external thing. But this is irrelevant to the relation involved in *zhengjie* (Jin 2010b: 99, 123). For a similar reason, Jin did not accept the representational theory of knowledge. According to him, no matter what “representation” a particular type of representational theory refers to, talking about the representation of the external thing is to hold that the representation and the external thing that is represented are two things. Such a view implies that the existence of the external thing is indirectly inferred from the senses, while his view on *zhengjue* is about directly and correctly sensing the external thing. Therefore, *zhengjue* is not a representational relation (Jin 2010b: 99–100, 122). But, how can we know when we have *zhengjue* or correct sense perception of things? Again, Jin appealed to common sense. He seemed to say that we all believe by our common sense that we know when our senses function normally and perceive things correctly. Furthermore, we all believe that we sense the external thing itself when we have *zhengjue* (Jin 2010b: 93).

The second step of knowing is to process materials for knowledge provided by senses and abstract universal meaning from our sensing of perceived particular and concrete things. This is where abstraction plays a key role in the formation of knowledge. Jin described the process of abstraction by saying we “use one to measure many and use a model to measure actuality” (*yi yi fan duo* 以一范多, *yi xing fan shi* 以型范實). This process consists of two steps. According to Jin, the first step of abstraction is to form a certain image (*yixiang* 意象) that can serve as a model or symbol to identify individual objects in a certain kind of thing. For example, when a person sees a train for the first time, she may form an image of the train and regard the image as a representative of trains, and therefore use the image to identify individual trains. At this stage, the “one” that serves as a criterion to identify many particular objects is still an image (Jin 2010b: 168–69; Wang 2004: 158–59; Wu 2013: 326). Abstraction has started in taking an image as a symbol or model, but an abstract idea that has no images has not been formed. In the next stage, an abstract idea or concept is formed by thinking about images or materials acquired in the previous stage. This is when the formation of abstract ideas (*yinian* 意念) or concepts takes place. Ideas that are formed at this stage constitute knowledge that has some universal meaning. Returning to the example of knowing what a train is, after one sees a train and takes what she sees as a symbol for a train, if she wants to tell others what a train is, besides drawing a picture of what she sees, she may describe it by using a series of concepts that she has known. By doing so, her image of train jumps to her conception of train, i.e., she forms a concept of train or a definition of train (Jin 2010b: 169; Wang 2004: 159).

According to Jin, ideas or concepts derived from abstraction play a very significant role in knowing. Among other things, they may function in the following two ways. First, they symbolically arrange what is presented in sensation in the whole picture of ideas so that what it means in a particular and concrete content can be preserved and passed on. He called this function “description of particular things”

(*mozhuang* 摹状) (Jin 2010b: 260). Second, they are used to receive and arrange new particular and concrete things presented in sense perception by classifying them according to concepts. He called this “regulation” (*guilü* 规律) (Jin 2010b: 266; Wang 2004: 164–165).

More specifically, *mozhuang* is a process of using existing concepts to conceptualize particular objects that an epistemic agent perceives. This involves a process like this: first one puts what one perceives into some known categories and then uses these known concepts to describe what is unknown; and finally one puts the unknown into a new category. For example, when one perceives a particular and unfamiliar color from an object, by first identifying other colors close to it via familiar concepts of colors, one will be able to describe it in terms of known concepts of colors, such as defining it as something between two known colors that she has known. In this way, she can label it by a concept of a particular color. For instance, if she perceives two colors in an object as yellow and red and an unknown color locates between yellow and red in the color spectrum, she would be able name the unknown color as red-yellow or yellow-red and therefore call it “orange.” Then, this particular color obtains a definition and her sense perception of this particular color will be expressed to others in a universal term, and therefore knowledge of this particular color will be formed and able to be passed on (Jin 2010b: 264–265). In general, *mozhuang* is about using existing concepts to describe particular things or qualities that are not in existing concepts and therefore form new concepts. This process heavily depends on experience of particular things. *Guilü* clearly functions differently from *mozhuang*, since it is about putting coming contents received in sense perception into existing concepts. As the way to accept things into certain categories, it mainly relies on existing concepts to classify and identify things (Jin 2010b: 266–67). For example, when one sees some particular fruits, one will use the definitions of apple, pear, banana, or peach, etc., to accept or put these particular fruits to particular kinds of fruit according to the existing definitions (Wang 2004: 164–165). This is to apply general ideas or concepts to particular objects. According to Jin, a definition or concept of something is like a law that strictly defines what object can be accepted to a class of thing. A concept can’t regulate how things can be presented in our experience, but it decides how things will be accepted into certain categories (Jin 2010b: 267–268). However, for Jin, *mozhuang* and *guilü* can’t be separated. Without *guilü*, *mozhuang* can’t be done since it has to use and create concepts to classify things, while *guilü* can’t be done without *mozhuang* since it must identify individual cases by experience for the classification of things. For the sake of analysis, they can be regarded as two components of concepts but can’t be two actual processes in reality (Jin 2010b: 280).

True Propositions and the Criteria of Correspondence

According to Jin, the discussion of knowledge is about truth (Jin 2010b: 650). His *Epistemology* is a book about truth/falsehood (Jin 2010b: 55). However, he confined his investigation of truth to the truth and falsity of propositions. Furthermore,

propositions that his epistemology focuses on are synthetic but not analytic (Jin 2010b: 55). Before we proceed to a discussion of Jin's view of the criterion of truth, it is necessary to explain Jin's distinction between "Truth" as the whole and "truths" of propositions (for the sake of convenience the former in this chapter is expressed as "Truth" with a capital letter). For Jin, "Truth" refers to the overall structure of all true propositions. It is the ultimate goal and limit of our knowledge. We can never reach it but can always be motivated by it (Jin 2010b: 697; Wu 297–98). However, we can make constant progress in obtaining truths of propositions. Epistemology studies truths of propositions but not Truth as the unachievable goal. We can never claim that we have obtained total truths of all propositions, therefore we can never achieve Truth.

Then, when is a proposition true? Among existing and possible theories of knowledge, Jin thought that the correspondence theory of truth is best. According to him, a proposition is true if there is the correspondence between the proposition and the reality or fact that the proposition states (Wu 290–291). To better support his view, he first discredited three other theories of truth, i.e., the doctrine of harmonization (*rongqia shuo* 融洽說), the doctrine of effectiveness (*youxiao shuo* 有效說), and the doctrine of consistency (*yizhi shuo* 一致說) (Chen 2021a: 43–44). Before Jin proceeded to criticize these three theories, he clearly acknowledged that they might not be exactly the theories that are actually held by any philosopher and that in each of them there is something valuable which could be incorporated in his own epistemology (Jin 2010b: 650). According to Jin, the doctrine of harmonization takes empirical and comprehensive harmonization as "true." In this view, truth requires the harmonization of all parts and structures in the whole (Jin 2010b: 650–651). All in the whole are organically related together and internally connected. Truth requires completion and wholeness. The truth of parts is incomplete. It is true only to some degree but not fully. As far as propositions are concerned, this theory implies that truth involves the entirety of all propositions that are in harmony and organically connected together. The truth of the whole entails the truth of parts, but not the other way around. The complete truth in this doctrine is like what Jin called "Truth" as the ultimate goal and limit of knowledge (Jin 2010b: 651). For Jin, this doctrine of truth is too metaphysical to fit his theory of knowledge. He emphasized that what his epistemology concerns is ordinary knowledge in daily life and what he is interested in is truth/falsity of propositions (Jin 2010b: 651–652). The doctrine of effectiveness is close to what pragmatism or experimentalism believes about truths, which holds that assumptions or hypotheses are true if they are useful and feasible in practice. But it fails to see that not all effective assumptions are true although true propositions are indeed useful and feasible. Furthermore, given that effectiveness is relative to a particular time and place and that nothing is permanently effective, to equate truth with effectiveness is to make truth relative and temporary, which is not what Jin could accept. The doctrine of consistency holds that truth lies in consistency between propositions, but not in correspondence between proposition and fact. For Jin, this theory can't provide the right criterion to judge whether a proposition is true since one of two inconsistent propositions may be true (Jin 2010b: 654). More importantly, the consistency that this doctrine advocates

only involves the relationship between propositions, which has nothing to do with truth or falseness of a proposition. Whether a proposition is true or false is determined by the relation between a proposition and the external fact that the proposition states (Jin 2010b: 654).

Jin acknowledged that the correspondence theory of truth does have its problems. First, it faces some difficulty in explaining in what sense a proposition can correspond to a fact, given that a proposition is not like a photo that obviously can correspond to the object in the photo. Second, it faces even a greater difficulty in explaining how the epistemic agent can know whether a proposition indeed corresponds to a fact. Since judging whether something corresponds to something else requires knowing both and comparing both but the epistemic agent may not know the fact that the proposition needs to correspond prior to judging whether a proposition correspond to the fact, she may not know whether a proposition indeed corresponds to a fact (Jin 2010b: 655–656). Nevertheless, Jin insists that we shouldn't give up the correspondence theory because of these difficulties (Jin 2010b: 666). Why should we adopt the correspondence theory according to Jin? First, it is close to common sense (Jin 2010b: 655; Wu 291). In our daily life, we mostly take correspondence as the definition of truth. Since common sense admits that there is an external world independent of the epistemic agent and propositions are usually statements about the external world, the truth of propositions must be related to the external world and true propositions correspond to it. Although common sense is not always correct, it can't be all wrong. As pointed out earlier, for Jin, ultimately, the errors of common sense have to be corrected by common sense (Jin 2010b: 657). Any discipline must start with or be based on common sense to some degree. Without common sense, no scholarship can even begin (Jin 2010b: 655–656; Chen 2021a: 46). Second, in practice, the correspondence between propositions and external realities is actually taken as the definition of truth even by those who theoretically reject it (Jin 2010b: 656; Chen 2021a: 46–47). Furthermore, taking such correspondence as the definition of truth can actually work since the gap between a proposition and a fact or reality can be bridged. Given that *zhengjue* directly perceives external objects and that perceivers use concepts to accept and then classify and arrange the given, the subject and the object are connected (Chen 2021a: 47). Jin didn't work out more details on this, but he did make efforts to connect two sides in the process of knowing (Chen 2021a: 47). However, to explain how correspondence can work as the definition of truth, Jin needs to show what kind of correspondence is required for the truth of a proposition. According to him, the correspondence in this context can't be the one between a photo or a copy and its original (Jin 2010b: 655). The correspondence between a proposition and a fact is structural, which is similar to the one between locations in a map and actual locations, or the one between a library catalog and the books in the library. It may be like what Russell called "the parallel structure" between language and the world (Jin 2010b: 671–672; Chen 47).

How can we know when such correspondence occurs between a proposition and reality? It seems that we can't know if there is a correspondence between the two unless we first know the way the reality is, but we can't know that directly since we

can only know the reality external to us by what we receive from the external world through our senses. Then, we can't guarantee what we perceive is exactly what it is external to us (Wu 302). Even if the truth of a proposition is indeed the correspondence between the proposition and the reality, it seems that there is no direct way for us to know when such correspondence occurs. There seems to be a gap between proposition and reality which Jin claimed to be bridged by correct sensation in which external things are directly perceived by the epistemic agent. However, Jin's argument on that doesn't sound very compelling. As has been pointed out, Jin didn't resolve the issue at a theoretical level or work it out in detail (Hu 2002b: 373; Chen 2021a: 47). Jin simply asserted that *zhengjue* perceives the way external things really are but didn't have proof except for claiming that our common sense believes so. For him, no matter what difficulty the correspondence theory faces, people's common sense accepts the view that truth consists of correspondence. But whether common sense actually accepts such a view in itself is questionable (Wang 2004: 176).

However, Jin did propose three necessary conditions for the correspondence between a proposition and a fact: (1) harmonization (*rongqia xing* 融洽性), (2) effectiveness (*youxiao xing* 有效性), and (3) consistency (*yizhi xing* 一致性). *Rongqia xing* is the harmony between sensation and what is sensed, which has nothing to do with interconnectedness (*rongqia* 融洽性) in the doctrine of *rongqia*). In short, "*rongqia* is the felt harmonization between a proposition and its object" (Jin 2010b: 680). In other words, it is the epistemic agent's feeling of harmonization between a proposition and the reality it refers to. When one feels disharmony between two, a proposition can't be true. For example, when one put his feet into water to wash but yelled because water either is too cold or too hot, he felt that his assumption that water was not too hot or too cold was not in harmony with the fact that water was too hot or too cold. What he assumed (i.e., the proposition he believed) was not in harmony with what he perceived as reality. Then, it was indicated that his assumption was not true (Jin 2010b: 674). Effectiveness is a characteristic of truth as well. In this context, "effectiveness is the effectiveness of a proposition in action, movement, or outcome" (Jin 2010b: 680). What is true must be effective, although what is effective is not necessarily true (Jin 2010b: 679). Such understanding of the relation between effectiveness and truth distinguishes Jin's view from the doctrine of effectiveness according to which effectiveness is a sufficient condition of truth. As far as consistency as a condition for truth in Jin's theory is concerned, it is not simply consistency among propositions but consistency among true propositions. According to Jin, all true propositions must be consistent, although consistent propositions are not necessarily true (Jin 2010b: 680). Harmonization, effectiveness, and consistency are all necessary but not sufficient conditions for correspondence (Jin 2010b: 682; Chen 2021a, b: 47–48). Therefore, Jin claimed that correspondence between proposition and reality is the definition of truth, while harmonization, effectiveness, and consistency are the criteria of correspondence (Jin 2010b: 682; Chen 2021a, b: 47). Furthermore, the epistemic agent will feel the sense of correspondence when a proposition meets all these criteria of correspondence or necessary conditions for its truth. For Jin, harmonization and

effectiveness are always associated with feeling or sense, which are empirical and linked to particular times, but consistency is not since it only deals with the relation between propositions. With progress of knowledge, harmonization and effectiveness as the criteria of correspondence will become more and more accurate and delicate and people's feelings of harmonization and effectiveness will become more exquisite. In this sense, "later comers are better" as far as these two are concerned. What people felt to be harmonious and effective in the past might be no longer felt the same today. They evolve with time (Jin 2010b: 681). But consistency doesn't change in time. What is inconsistent is always inconsistent, even if we didn't find it out in the past or don't feel it (Jin 2010b: 681–682). Jin obviously tried hard to make correspondence theory work by proposing these criteria of correspondence, but his effort didn't fully succeed, although it was insightful in some ways. Making the two of three criteria of correspondence rely on so much on sensory experience, which is relative and subjective to a great degree, he was not able to show that epistemic agents can ever truly know whether a proposition objectively corresponds to the reality that it refers to. Of course, we may regard his three criteria of correspondence as the result of the analysis of common sense that takes correspondence between propositions and realities as truth.

The Permanent Validity of the Principle of Induction

Jin believed that the principle of induction is always true as long as human experience continues (Jin 2010b: 310). For him, the principle of induction is a necessary condition for experience without which no one can have experience of the world. It is in this sense that Jin called the principle of induction "a prior principle." Furthermore, he believed that the principle of induction is also the general way for people to accept and process the given. When the mind accepts and deals with the given, it must use the principle of induction (Jin 2010b: 335–337; Wang 2004: 189).

As mentioned earlier, Hume's challenge to inductive reasoning was one of the most important factors that initially got Jin interested in philosophy. Defending the principle of induction was part of Jin's motivation to devote himself to philosophy. According to Jin, the principle of induction enables us to universally infer universal propositions from particular instances. Although conclusions of different inductions may be different in their application scope, they follow the same principle of induction (Jin 2010b: 312). Usually, induction is about inferring universal propositions from particular instances observed empirically up to the present. Hume pointed out that induction as such can't logically support necessarily and universally true conclusion since all that is empirically observed occurred in the past and present and doesn't entail what will happen in the future. To say that something is universally and necessarily true means that it is always true, including in the future, without exception. Unless we know that the future and the past are uniform or very similar, we have no reason to conclude that what has happened will necessarily happen in the future. But we can't logically or empirically prove the uniformity of the past and

the future. Therefore, universal propositions inferred from induction such as “fire produces warmth” aren’t logically necessary, and our belief in them is not rational but arises out of habit or psychological association between two events that have been constantly conjoined in our experience. Jin well understood what Hume’s problem of induction is and was bothered greatly by it. In his epistemology, unlike in his metaphysics, Jin didn’t intend to directly challenge Hume’s question on universality and necessity of inductive truth but focused on the permanent validity of the principle of induction.

As Jin expresses it, suppose $a_1, b_1, a_2, b_2, a_3, b_3 \dots a_n, b_n$ are used to represent particular things or events, $a_1 \rightarrow b_1, a_2 \rightarrow b_2, \dots a_n \rightarrow b_n$ are used to represent past and present instances of the connection between a and b , A and B are used to represent classes of a and b , and $A \rightarrow B$ the connection of a and b , the principle of induction may be formulated as follows: if $a_1 \rightarrow b_1, a_2 \rightarrow b_2, \dots a_n \rightarrow b_n$, then (probably) $A \rightarrow B$ (i.e., a member in the class A will probably be always followed by a member in the class B) (Jin 2010b: 311).¹² For instance, if in the past (up to now) fire was always followed by warmth, then fire will probably be always followed by warmth.¹³ Obviously, this formulation of the principle of induction is not exactly the same as the original one that Hume questioned, since “probability” is added, which makes the principle become the one merely about probable but not necessary truth by induction. Then, even if there are some counterexamples for a universal proposition derived from induction, the principle of induction will not be falsified. As Jin said: if we use the concept “probability,” we indeed can say that if part is true, then the whole is probably true. The principle of induction is such a proposition (Jin 2010b: 325). The modified definition of the principle of induction entails Jin’s acceptance of Hume’s view that induction doesn’t give necessary truth. However, adding “probably” to the principle of induction can’t really solve Hume’s problem of induction, since past instances of the connection between a and b even can’t guarantee any occurrence (even a merely probable one) of the connection between a and b in future. What happened in the finite past doesn’t imply any probability of what will happen in the infinite future (Chen 2001).

As a matter of fact, Jin claimed that he didn’t think “probable” is important for his discussion of the principle of induction. What he focused on in his epistemology was why the principle of induction is always true or permanently valid by deductive arguments. He has made two arguments in this direction. First, he argued that the principle of introduction as “ $P \rightarrow Q$ ” (if ... then), a conditional statement, is always true except when P is true and Q is false, but it is impossible for the principle of induction to have false Q and true P . P as the antecedent represents a statement on all observed instances of $a \rightarrow b$, e.g., “fire is always followed by warmth up to this point,” and Q represents the consequent, the universal proposition based on P ,

¹² Jin acknowledged that this formulation is based on Russell’s summary of the principle of induction (Jin 2010b: 307, 310–311). For Russell’s formulation of the principle of induction, see Russell (1945: 673).

¹³ Jin didn’t use this example. For the sake of convenience, I use it to illustrate his formulation and related points in this section.

namely, “ $A \rightarrow B$ ” (for all a and b , $a \rightarrow b$), e.g., “fire is followed by warmth all the time (including in future).” If both P and Q are true, $P \rightarrow Q$ is certainly true; if both P and Q are false, $P \rightarrow Q$ is still true. Based on this, Jin claimed that $P \rightarrow Q$ as the principle of induction is always true, since it is impossible for it to have $P \& \neg Q$. Here Jin seems to make two logical errors. First, in “ $P \rightarrow Q$ ” as the principle of induction, whether Q is true may be unknown. That is what Hume’s problem of induction has clearly indicated. Therefore, Jin shouldn’t have concluded that $P \rightarrow Q$ is true because both P and Q are either both true or both false. When Q may be unknown, the truth value of $P \rightarrow Q$ is unknown. Second, to claim that it is logically impossible for $P \rightarrow Q$ to have $P \& \neg Q$ presupposes that $P \rightarrow Q$ is true and P is a necessary and sufficient condition for Q . Otherwise it is logically possible that $P \& \neg Q$ and then $P \rightarrow Q$ is false. In the case of the principle of induction, its antecedent P (in the past it is always the case that $a \rightarrow b$) doesn’t logically entail Q (i.e., at all times, including in the future, it will always be the case that $a \rightarrow b$). To assume that $P \rightarrow Q$ is true and P is a necessary and sufficient condition for Q in this case is to use the principle of induction to prove the principle of induction (Chen 2001). Without realizing his logical errors, Jin went further to argue that P and not- Q will be possible in the case of $P \rightarrow Q$ as the principle of induction only if time stops. According to Jin, when time stops, Q will become false in the sense that it becomes a conclusion of past instances but not a universal proposition that predicts the future instances. However, since time will not stop and there will always be future, Q will not become as such. Therefore, “ $P \& \neg Q$ ” will never occur in $P \rightarrow Q$ as the principle of induction, which guarantees the permanent validity of the principle of induction. But, as Wang Zhongjiang points out, even if time stops, that merely means that the principle of induction will no longer function, but not that Q is false while P is true if P is a necessary and sufficient condition for Q (Wang 2004: 187).¹⁴

Another argument that Jin offered for the permanent validity of the principle of induction focuses on demonstrating why the principle of induction can’t be falsified by future instances. Without being able to prove in the future “if a , then b ” ($a_{n+1} \rightarrow b_{n+1} \dots$), how could he claim that the principle of induction is forever true? Jin argued that any future counterexamples for a universal proposition inferred from induction don’t falsify the principle of induction itself but merely falsify a proposition that is the outcome of the application of the principle of induction. According to Jin, when one uses the principle of induction to get a conclusion, one actually has two premises: (1) the principle of induction as the first premise, and (2) particular instances as the second premise. Which universal proposition as a conclusion one will get from a specific inductive reasoning is determined by the second premise (Jin 2010b: 319). If there are counter examples for a particular universal conclusion from induction, what is falsified is merely that particular conclusion but not the first premise, i.e., the principle of induction itself (Jin 2010b: 319). For example, if there are counterexamples for “fire is followed by warmth” in the future, they can only

¹⁴ Neither Chen Bo nor Wang Zhongjiang uses “ $P \rightarrow Q$ ” to represent the principle of induction. It is my own expression. But some of their ideas fit well when $P \rightarrow Q$ is used.

falsify the conclusion that “fire is followed by warmth and therefore produces warmth all the time,” but don’t falsify the principle that if all instances so far show $a \rightarrow b$, then it is probably $A \rightarrow B$ in the future. He counted a particular future counterexample “ $\neg (a_{t_{n+1}} \rightarrow b_{t_{n+1}})$ ” as an instance of “fire is not followed by warmth” as part of the second premise of this inductive reasoning and therefore derived a specific universal proposition “ $\neg (A \rightarrow B)$ ” such as “it is not the case that fire always produces warmth.” He insisted that in this case the universal proposition “fire produces warmth” is falsified due to counter examples, but the principle of induction itself is not falsified (Jin 2010b: 324). His reasoning seems to be this: according to the principle of induction, probably $A \rightarrow B$ if all instances of the connection between a and b are $a \rightarrow b$; not probably $A \rightarrow B$ if some a and b are not connected as $a \rightarrow b$; when “not probably $A \rightarrow B$ ” is inferred from “some a and b are not connected as $a \rightarrow b$ ”, the principle of induction is clearly not false. The principle is false only if its antecedent is true and its consequent is false. But that is not what happens here. In the case of the proposition “fire probably is always followed by warmth,” according to the principle of induction, it is true if in all instances of the connection between fire and warmth fire is always followed by warmth; it is not true if some of its instances (including its future instances) show otherwise. A logical gap in this reasoning is obvious: “all instances” in the antecedent of the principle of induction don’t include future instances, while in Jin’s specific application of the principle (e.g., fire probably always produces warmth), “all instances” includes future instances. To bridge the gap, Jin needs to adjust the principle of induction so that it includes future instances in its antecedent. That seemed to be what he did. According to Jin, given that time goes without stopping and “now” or current time t_n will be the past and the future time t_{n+1} will be “now,” and any future instance must have become current if it could even falsify any universal proposition from induction, we may let the principle of induction include future instances in its antecedent by changing the stopping point of instances that induction is based on from now (t_n) into a future moment (t_{n+1}), then the principle of induction may be regarded as “if $a_1 \rightarrow b_1, a_2 \rightarrow b_2, a_3 \rightarrow b_3 \dots a_n \rightarrow b_n$, and $a_{t_{n+1}} \rightarrow b_{t_{n+1}}$, then probably $A \rightarrow B$ (Jin 2010b: 322). Unfortunately, Jin’s doing so is clearly illegitimate, as pointed out by Chen Bo (2001). Future and present can’t be exchanged because what is future now will not be future tomorrow. Predicting future occurrences from past regularities is the key to induction. He might be right in believing that future counterexamples against certain propositions from induction don’t falsify the principle of induction itself, but he didn’t realize that he couldn’t succeed in deductively demonstrating the truth of the principle of induction without using “the past and the future are similar” as a premise in his argument. However, in his metaphysics, he seems clear about this and made a great effort to demonstrate the uniformity of nature and the existence of natural laws.

According to Jin, fundamentally, Hume’s errors lie in his skepticism about universal laws or orders of the world and the uniformity of nature. As discussed earlier, Jin believed that all individual instances in nature are the manifestations of universal

laws or what Jin called “intrinsic connections of universals” (*guran zhi li* 固然之理) but not simply random occurrences. Individual instances as particular individuals express something universal that are always valid across different times, and therefore, those past instances used in inductive generalizations are not merely about what happened in the past but also about what will happen again in the future. However, Jin didn’t realize that the uniformity of nature may not be inferred deductively but could be reasonably taken as the best explanation of past regularities. Indeed, from past instances we can’t deductively infer what will happen in the future, no matter how many times they repeatedly occurred, but unexceptional repetitions of certain instances in the past such as “fire is followed by warmth” must be explained, and their best explanation could be that there is a necessary connection between *a* and *b* as the relation of cause and effect. Although Jin didn’t explicitly express such reasoning, his argumentation for universal laws, including the law of causality which is largely what induction relies on, may be developed in this direction. Although Jin constantly asserted that there must be universal laws or orders and that causality is universally manifested in particular things or events, he didn’t justify his belief well. If he could explain constantly repeated instances in terms of some hidden natural order or laws, his arguments for the existence of universal laws in nature and his defense for induction probably would have been much more compelling.¹⁵

Overall, Jin’s philosophical system is vast and comprehensive. He might be too ambitious in establishing a philosophy that can solve so many important philosophical problems at once. Although there are some obvious flaws and difficulties in his philosophy, Jin did have some insights that are worth considering and developing. More importantly, he was among very few Chinese philosophers who joined the mainstream philosophical discussions in the West and regarded true philosophy as something belonging to humankind across cultural boundaries. He didn’t deny that there are different philosophies in different cultures, such as Chinese philosophy and Western philosophy, but he did emphasize that there are common philosophical issues, such as causality and induction, which are universal topics for all philosophers over the world. As a Chinese philosopher, he was bold and courageous to participate in global philosophical conversations more than anyone else before and after him in China. He established an example of what a Chinese philosopher could do in philosophy worldwide within one’s culture. His philosophy itself may be immature but his effort could be a great inspiration for later generations of Chinese thinkers to contribute to world philosophy in their own ways.

¹⁵ There are some attempts to solve the problem of induction in this direction in contemporary literature on the subject. For a brief discussion of these efforts, see the section, “3.2 The Nomological-Explanatory,” in Leah Henderson (2022).

References

- Chen, Bo 陳波. 2001. Hume's Problem and Jin Yuelin's Answer 休謨問題和金岳霖的回答. https://www.thepaper.cn/newsDetail_forward_15969949. Accessed 10 December 2023.
- Chen, Bo 陳波. 2021a. Russell and Jin Yuelin on Truth: A Comparative Study 羅素和金岳霖論真理: 一个比較研究. *Journal of Peking University* (Edition for Philosophy and Social Sciences) 北京大學學報哲學社會科學版 58:3: 34–53.
- Chen, Jiaming 陳嘉明. 2021b. Brief Discussion of Several Questions in Jin Yuelin's *Epistemology*. 略論金岳霖《知識論》中的幾個問題. <https://www.aisixiang.com/data/126866.html>. Accessed 16 December 2023.
- Guo, Qiyong 郭其勇. 2006. *A History of Chinese Philosophy* 中國哲學史. Beijing: Higher Education Press.
- Henderson, Leah. 2022. The Problem of Induction. In *The Stanford Encyclopedia of Philosophy* (Winter 2022 Edition), eds. Edward N. Zalta & Uri Nodelman. <https://plato.stanford.edu/archives/win2022/entries/induction-problem>. Accessed 18 February 2024.
- Hu, Jun 胡軍. 2002a. Jin Yuelin's Theory of *Dao*. In *Contemporary Chinese Philosophy*, eds. Chung-ying Cheng and Nicholas Bunnin, 102–123. Oxford: Blackwell Publishers.
- Hu, Jun 胡軍. 2002b. *Dao and Trueness: The Study of Philosophical Thought of Jin Yuelin* 道與真——金岳霖哲學思想研究. Beijing: People's Press.
- Hu, Jun 胡軍. 2004. On Tao 論道. In *Studies of Jin Yuelin's Thought* 金岳霖思想研究, ed. Peiyu Liu 劉培育, 2–124. Beijing: Chinese Social Sciences Press.
- Jin, Yuelin 金岳霖. 1998a. Chinese Philosophy 中國哲學. In *Chinese Philosophy Classics: The Volume for the Period of the 1930s and the 1940s* 中國哲學經典:三十、四十年代卷, eds. Yijie Tang 湯一介 and Wei-ming Tu 杜維明, 273–285. Shenzhen: Hai Tian Press.
- Jin, Yuelin 金岳霖. 1998b. Introduction to *Epistemology*. In *Chinese Philosophy Classics: The Volume for the Period of the 1930s and the 1940s* 中國哲學經典:三十、四十年代卷.
- Jin, Yuelin 金岳霖. 2009. A Report on Feng's *A History of Chinese Philosophy*. In Youlan Feng 馮友蘭. *A History of Chinese Philosophy* 中國哲學史, vol. 2, 459–462. Beijing: People's Press.
- Jin, Yuelin 金岳霖. 2010a. *On Dao* 論道. Beijing: China People's University Press.
- Jin, Yuelin 金岳霖. 2010b. *Epistemology* 知識論. Beijing: China People's University Press.
- Jin, Yuelin 金岳霖. 2022. *Jin Yuelin's Memoir* 金岳霖回憶錄. Beijing: Peking University Press.
- King, Hugh R. 1956. Aristotle Without Prima Materia. *Journal of the History of Ideas* 17 (3): 370–389.
- Liu, Peiyu 劉培育. 2004a. Comprehensive Discussion 綜合論. In *Studies of Jin Yuelin's Thought* 金岳霖思想研究, ed. Peiyu Liu, 341–396. Beijing: Chinese Social Sciences Press.
- Liu, Peiyu 劉培育. 2004b. Jin Yuelin Annals 金岳霖年表. In *Studies of Jin Yuelin's Thought* 金岳霖思想研究, ed. Peiyu Liu, 399–432. Beijing: Chinese Social Sciences Press.
- Russell, Bertrand. 1945. *A History of Western Philosophy*. New York: Simon & Schuster.
- Wang, Zhongjiang 王中江. 1993. *Rationality and Romance: Life and Philosophy of Jin Yuelin* 理性與浪漫: 金岳霖的生活及其哲學. Zhengzhou: Henan People's Press.
- Wang, Zhongjiang 王中江. 2004. The Theory of Knowledge. In *Studies of Jin Yuelin's Thought* 金岳霖思想研究, ed. Peiyu Liu, 127–221. Beijing: Chinese Social Sciences Press.
- Wu, Rujun 吳汝鈞. 2013. *Epistemology in Contemporary Chinese Philosophy* 當代中國哲學的知識論. Taipei: Publishing Center of National Taiwan University.
- Xu, Qianxing 許全興, Chen, Zhannan 陳佔南, and Song, Yixiu 宋一秀. 1992. *A History of Contemporary Chinese Philosophy* 中國現代哲學史. Beijing: Peking University Press.